Predicting Antimicrobial Resistance Without Local Data: Zero-shot country-level transfer to Kazakhstan

Uais Amangeldi, Dinmukhammed Amantay, Dauren Ayazbayev, Serikzhan Kazi

June 5, 2026

1 Abstract

In high-income nations, the use of machine learning models for antimicrobial resistance prediction has demonstrated remarkable performance; however, such models cannot be implemented in areas without local surveillance systems. With 1.14 million deaths attributed to AMR in 2021, countries such as Kazakhstan have limited access to AMR data to inform empirical treatment, due to the absence of a national surveillance system. Using 11 macroeconomic indicators, antimicrobial use indicators, and knowledge about the type of organism and antibiotic class, in this study a zero-shot transfer learning paradigm that assigns calibrated probability values is presented, based on a pre-trained CatBoost classifier trained on 20.4 million antimicrobial isolates from 85 countries. AUC, recall, and ECE measures were used to assess the method through leave-one-country-out cross-validation, regional modeling by cluster, and temporal hold-out testing. The model was able to achieve an AUC of 0.8558 with a recall of 0.7761 at the global level, while a model built using only 2.5% of the data achieved an AUC of 0.8983. As for implementation of the approach in Kazakhstan, it was revealed that in 2017–2023, Enterococcus faecium and Acinetobacter baumannii were in the high-risk category, with 54.6% of inferences above 20%. This finding indicates that macro-feature transfer learning can be used as a risk-prioritization tool for data-poor countries, helping to flag which organism–antibiotic combinations warrant priority testing, pending validation with local isolates.

2 Introduction

Antimicrobial resistance (AMR) poses a significant public health issue, with 4.71 million AMR-attributed deaths, and 1.14 million direct deaths caused by AMR in 2021 (Naghavi et al., 2024). Without proper interventions, annual

AMR cases can reach 10 million by 2050 (O'Neill, 2016), with estimated annual GDP losses of 3.4 trillion USD globally up to 2030 in the case of a high prevalence of AMR (World Bank, 2017). Even though AMR is a global problem, its impact is not evenly distributed and its effects are worse in LMICs due to a lack of appropriate laboratory diagnostics and infection control capacities (Laxminarayan et al., 2013). Isolates represented in global surveillance databases such as WHO GLASS, ATLAS, and SIDERO-WT are mainly isolated in high-income countries, with resistance profiles in LMICs remaining significantly underrepresented. Central Asia constitutes another example in which a large proportion of carbapenem-resistant Enterobacteriaceae among hospital-acquired infections in Kazakhstani, Kyrgyzstani, and Uzbekistani patients has been recorded (Semenova et al., 2024); however, no nationwide AMR surveillance program is available for empirical investigation. To address this surveillance gap, we explore whether a model trained solely on global data can predict resistance in a country that contributed no training isolates—an approach we refer to as zero-shot transfer, denoting a model applied to Kazakhstan without any local data contributing to training or validation. As shown by previous phenotypic prediction methods, epidemiological and pharmacological factors have the potential to predict AMR even in the absence of genomic information (MacFadden et al., 2018; Pires et al., 2021). However, all existing methods of validation assume that isolates from the region of study are available, ignoring data-scarce countries such as Kazakhstan. In settings where no local susceptibility data exist, minimizing false negatives—resistant isolates incorrectly labelled as susceptible—is clinically paramount, since a missed resistant infection carries greater risk than a false alarm. We therefore address the surveillance gap using a zero-shot transfer approach in which a CatBoost classifier is trained on global data and then applied to Kazakhstan using only publicly available macroeconomic indicators and antibiotic consumption data as input features. This work contributes in three ways. First, to our knowledge, this is the first application of zero-shot AMR prediction to a Central Asian country, specifically Kazakhstan. Second, we provide three validation strategies that do not depend on local ground truth: temporal splitting, Leave-One-Country-Out (LOCO), and rolling temporal cross-validation. Third, we demonstrate that readily available macroeconomic and antibiotic consumption data can be used to prioritize AMR testing in data-sparse settings, offering a hypothesis-generating framework that is potentially transferable to other under-surveilled regions. We emphasize that this framework is not meant to replace laboratory-based surveillance, as no national isolate-based system exists in Kazakhstan, and it is intended only as a risk-prioritisation tool that flags which organism–antibiotic combination should be tested first. We hypothesize that a model trained on global surveillance data, when applied to Kazakhstan using only macroeconomic and antibiotic consumption features, will still produce resistance probability estimates exceeding the 20% clinical actionability threshold for high-risk organism–antibiotic combinations, demonstrating that macro-level signals are sufficiently informative for risk prioritization in the absence of local surveillance data.

3 Methodology

3.1 Study Design

The current study presents a zero-shot country-level transfer learning framework for predicting the AMR pattern in Kazakhstan. The proposed pipeline design and process are detailed in Figure 1, which involves a global AMR surveillance database along with country-level macroeconomic and antibiotic consumption data. First, the process involves pre-processing, feature extraction, and modeling based on the global AMR surveillance dataset without including Kazakhstan data. The trained models are then tested on Kazakhstan, which serves as an unseen domain using only macroeconomic and antibiotic consumption data. For validation purposes, temporal splitting and cross-validation were done using three reference datasets, namely, two cluster-based datasets (post-Soviet and structural peers) and one global leave-one-country-out (LOCO) dataset that includes all countries except Kazakhstan. A sensitivity analysis was carried out to assess the robustness of the ensemble learning method under different weighting methods (equal weights, removal of the structural peers model, and performance-based weights). Resistance probabilities were estimated and temporal trends were analysed to determine high-risk organism–antibiotic pairs and emergent AMR patterns in Kazakhstan.

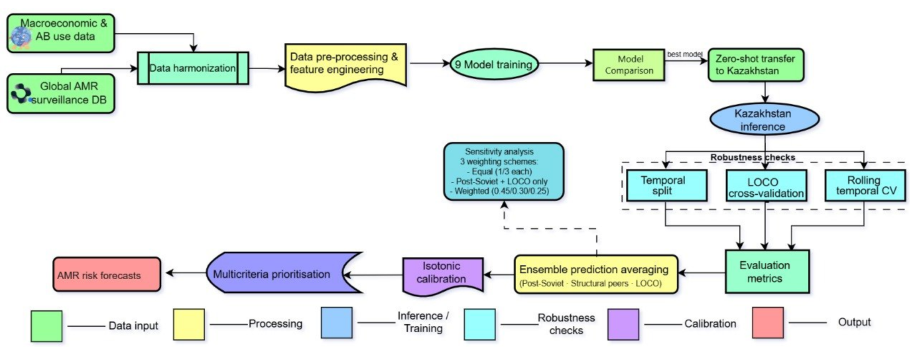

Figure 1: *Overall workflow of the machine learning pipeline from data preprocessing to prediction.*

3.2 Data

3.2.1 Data Sources and Acquisition

The dataset consisted of two components defined by their role in the zero-shot framework: a global dataset, which served as the sole source of all model training, validation, hyperparameter selection, and probability calibration; and a Kazakhstan target dataset, reserved exclusively for zero-shot inference and post-hoc face validity, contributing no isolate-level labels to any stage of model fitting.

The global clinical dataset was compiled from isolate-level data in the ATLAS (Pfizer), SIDERO-WT (Shionogi), and KEYSTONE databases, obtained after submitting data request applications to http://vivli.orgVivli.org. The data on consumption of antibiotics (DDD per 1,000 inhabitants per day) were obtained from Wang et al. (2025). Because no national AMR surveillance system exists in Kazakhstan, no isolate-level susceptibility data were available for model training. For the Kazakhstan target dataset, all features were country-level proxy features in the same feature space as the training data. The national antibiotic consumption rates were combined from four published articles, including Zhussupova et al. (2021), Semenova et al. (2024), Kassym et al. (2025), and Lim et al. (2025). Six macroeconomic indicators — GDP per capita, health expenditure (% of GDP), population density, access to basic water, access to sanitation, and hospital beds per 1,000 (World Bank Open Data, 2017–2023) — together with annual mean surface temperature (Our World in Data) and WHO AWaRe Access percentages (Zhussupova et al., 2021; WHO, 2023), completed the Kazakhstan feature grid. These features were used only to generate resistance probability estimates at inference time; published Kazakhstani resistance rates were used solely for the independent face-validity comparison (Section 5.4.3) and never entered model training or calibration. Details on the complete data acquisition are outlined in Table 1.

Table 1: *Data Sources and Acquisition*

Category	**Source**
Global Datasets	ATLAS (Pfizer), SIDERO-WT (Shionogi), KEYSTONE
Acquisition	Vivli.org (formal request)
Kazakhstan Consumption Dataset	Acquired from merging Gulzira Zhussupova et al. (2021), Yuliya Semenova et al. (2024), Laura Kassym et al. (2025), Lisa Lim et al. (2025).
Macroeconomic Data	World Bank Open Data
Global Antibiotic Consumption Dataset	Wang et al. (2025)
Average Monthly Surface Temperature	Our World in Data
Access Percentage	WHO AWaRe; Zhussupova et al. (2021)

3.2.2 Data Harmonization

Consistency of column names was achieved by replacing them with standard names, and name inconsistencies across countries were addressed using a mapping table. Entries erroneously coded as antibiotic names were removed. Binarization was done with MIC thresholds, where EUCAST species-specific thresholds resulted in 99.93% label accuracy, thus confirming that antibiotic-level labels are adequate for model training (Supplementary Table S1). In cases where no organism-specific breakpoint was available, single conservative MIC cutoffs were applied (Supplementary Table S2). Intermediate category isolates were dropped to ensure the existence of a clear distinction between susceptible and resistant isolates. Names of antibiotics in the Kazakhstan dataset were converted to lowercase and mapped to the global set of antibiotics, while antibiotics that could not be mapped were treated as unseen classes (Supplementary Table S4). Assignment of antibiotic class was done according to Table 2.

Table 2: *Antibiotic-to-class mapping used for feature engineering.*

Antibiotic	**Class**
Meropenem	Carbapenems
Imipenem	Carbapenems
Doripenem	Carbapenems
Ertapenem	Carbapenems
Ciprofloxacin	Fluoroquinolones
Levofloxacin	Fluoroquinolones
Moxifloxacin	Fluoroquinolones
Ceftazidime	Cephalosporins
Cefepime	Cephalosporins
Ceftriaxone	Cephalosporins
Amikacin	Aminoglycosides
Gentamicin	Aminoglycosides
Ampicillin	Penicillins
Amoxycillin clavulanate	Penicillins
Piperacillin	Penicillins
Azithromycin	Macrolides
Erythromycin	Macrolides
Vancomycin	Glycopeptides
Linezolid	Oxazolidinones
Colistin	Polymyxins
Tigecycline	Tetracyclines
Trimethoprim sulfa	Sulfonamides

3.2.3 Feature Engineering

Features related to macroeconomics and consumption were combined with the AMR data based on a yearly country-level dataset. Features that exhibited right skewness, that is, when most countries have moderate values while a few wealthy countries have extremely high values, such as GDP and consumption data, were log-transformed with $\log(1 + x)$. Population density for the case of Kazakhstan was calculated manually using the population and surface area of the country, since no dataset was provided by the World Bank regarding this variable. Missing values were forward-filled using chronological order, while the missing values remaining after that were imputed using the average of the corresponding columns at a global level. The final feature set comprised 14 variables: 3 categorical (organism, antibiotic, antibiotic class) and 11 numeric (year and 10 macroeconomic features; Table 3). Across the final training dataset, forward-fill resolved the majority of missing macro-feature values; global mean imputation was required for a median of 6.4% of country-year observations per feature, with the highest missingness in aware access pct (58.0%, 53 countries requiring global mean) and consumption DDD per class (57.6%, 52 countries). Core World Bank indicators (GDP per capita, population density, health expenditure) showed the lowest missingness at 5.4%, affecting 10 countries consistently lacking World Bank reporting coverage.

Table 3: *Final feature set used for model training and inference.*

Feature Type	**Feature Name**
Categorical	organism
	antibiotic
	antibiotic_class
Numerical	year
	log_gdp_per_capita
	health_expenditure_pct_gdp
	population_density_per_sq_km
	log_consumption_ddd
	log_consumption_did_per_class
	sanitation_access_pct
	water_access_pct
	hospital_beds_per_1000
	mean_temp_celsius
	aware_access_pct

3.3 Zero-Shot Design

Kazakhstan was completely excluded from the processes of model training, validation, and early stopping. Models were pre-trained using global AMR data and then applied to Kazakhstan as an unseen area based solely on macroeconomic and consumption proxy variables. For Kazakhstan's inference grid, national data on antibiotic consumption per 1,000 population per day (2017–2023) were combined with data on six WHO priority bacteria (*E. coli*, *K. pneumoniae*, *S. aureus*, *P. aeruginosa*, *A. baumannii*, *E. faecium*), resulting in 1,098 instances (183 antibiotics $\times$ 6 pathogens). All features were processed in the same way as the global training dataset through imputation, logarithmic transformation, and antibiotic class classification (Sections 2.2.2–2.2.3).

Three groups of references were established for the validation analysis. The post-Soviet reference group comprised Russia, Ukraine, Belarus, Lithuania, Latvia, and Estonia (Semenova et al., 2024; Kosherova et al., 2025). Structural peers were identified by using country-level macroeconomic and consumption data as input and through k-means clustering for k = 6, weighted by relevance to AMR issues (see Supplementary Table S3). As a result, Kazakhstan clustered with

Qatar. However, since Qatar was absent from the global dataset, Turkey, Romania, Bulgaria, Hungary, and Poland were included as structural peers. Global LOCO models were trained on all 85 training countries except Kazakhstan.

3.4 Model Development

A total of nine algorithms were considered: Logistic Regression (Cox, 1958), Random Forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016), LightGBM (Ke et al., 2017), CatBoost (Prokhorenkova et al., 2018), NGBoost (Duan et al., 2020), Explainable Boosting Machine (Nori et al., 2019), AutoGluon (Erickson et al., 2020), and Multilayer Perceptron (Rumelhart et al., 1986). Class imbalance (resistant/susceptible ratio = 1:4.5) was handled using a positive class weight (scale_pos_weight) in the case of gradient boosting algorithms and balancing class weight for Logistic Regression, Random Forest, and EBM; the exact ratio was recalculated independently in every training sample. Strict temporal validation (training up to and including 2018, early stopping in 2019–2020, final assessment after 2020) was performed to avoid potential data leakage. Hyperparameters were optimized through manual tuning according to the validation AUC score, as presented in Table 4. Due to computational restrictions, hyperparameters were estimated beforehand using the full global dataset, leading to a slight upward bias in LOCO-AUC estimates. However, this bias is not expected to be significant since no country represents more than 5 percent of the total number of 85 countries used in the training dataset.

Table 4: *Hyperparameter configurations for all nine candidate models.*

Model	Key Hyperparameters	Values
Logistic Regression	solver, max_iter, class_weight	saga, 1000, balanced
Random Forest	n_estimators, max_depth, min_samples_split, n_jobs	300, 15, 5, -1
XGBoost	max_depth, learning_rate, subsample, colsample_bytree, min_child_weight, scale_pos_weight, rounds	7, 0.03, 0.8, 0.8, 5, computed, 3000 (ES=100)
LightGBM	max_depth, num_leaves, learning_rate, subsample, lambda_l1, lambda_l2, min_data_in_leaf, rounds, colsample_bytree, min_split_gain	6, 48, 0.025, 0.75, 1.0, 3.0, 60, 5000 (ES=150), 0.75, 0.02
CatBoost	depth, learning_rate, l2_leaf_reg, iterations	7, 0.03, 3, 5000 (ES=100)
NGBoost	n_estimators, learning_rate, distribution, early_stopping_rounds	1000, 0.03, Bernoulli, 50
EBM	interactions, learning_rate, max_bins, class_weight	5, 0.01, 256, balanced
AutoGluon	presets, eval_metric, time_limit	best_quality, roc_auc, 3600s
MLP	layers, dropout, batch_size, lr, weight_decay, patience	256→128→64, 0.3/0.3/0.2, 4096, 1e-3, 1e-4, 10

3.5 Evaluation Methods

3.5.1 General Evaluation Metrics

Model evaluation involved AUC (Area Under the Curve), AUPRC (Area Under the Precision-Recall Curve), recall, F1 score, Brier score, BSS (Brier Skill Score), and ECE (Expected Calibration Error). Recall and AUPRC were selected as co-primary metrics owing to the clinical implications of false negatives, i.e., resistant isolates labeled as susceptible, rather than false positives, because missing a resistant infection can result in a failure to treat that may propagate resistance further (He & Garcia, 2009).

3.5.2 Calibration Methods

Calibration was determined based on ECE and BSS. Isotonic regression was estimated using the validation cohort (2019–2020) and calibrated the test set and Kazakhstan inference cohorts. Calibration data were obtained almost exclusively from high-income countries, thus posing a domain shift when transferred to Kazakhstan, which could be mitigated by the post-Soviet cluster where the validation dataset had greater similarities with the Kazakhstan macro-feature space.

3.5.3 Pairwise Model comparison method

Twenty-eight paired comparisons of AUCs were performed between the nine candidate models using the DeLong test (DeLong et al., 1988). Benjamini-Hochberg FDR correction was used as the main procedure: raw p-values were assembled as a family consisting of 28 p-values (one for each test), and each p-value was adjusted (corrected) for FDR using the standard step-up procedure: the p-values were ranked; each p-value was compared with the critical value $((i/28) \times 0.05)$, and significance was declared based on the FDR-corrected p-value at $\alpha = 0.05$. For completeness, Bonferroni-adjusted p-values are reported in Supplementary Table S6, although the Bonferroni correction is rather conservative given the large sample size.

3.5.4 Validation Methods

Geographic generalization testing comprised LOCO cross-validation (with early stopping in 2019–2020), as well as four-country hold-out testing (with Russia, Turkey, France, and India being entirely removed from the training and early-stopping process and tested with the held-out 100% dataset of each country).

The robustness of the ensemble approach was measured by conducting the analysis under three different weighting schemes: two-model averaging with equal weights (post-Soviet + global LOCO, 1/2 for each, i.e., weights = 0.5, 0.5); three-model averaging with equal weights (post-Soviet, global LOCO, structural peers, 1/3 each, weighting = 0.33); and weights based on model performances (post-Soviet = 0.45, global LOCO = 0.30, structural peers = 0.25).

3.5.5 Feature analysis methods

The PredictionValuesChange metric from CatBoost was used to calculate feature importance, and SHAP values were calculated using a stratified test set (n = 5,000) and TreeExplainer (Lundberg & Lee, 2017). If features are correlated in time, the two measures may yield different rankings: PredictionValuesChange measures the change in all predictions, and SHAP measures the contribution of each feature at a marginal level. The significance of the temporal dynamics of Kazakhstan's probabilities was tested by Spearman rank correlation with annual organism-level averages for 2017–2023, and by the Benjamini-Hochberg method (false discovery rate) with simultaneous testing of 6 probabilities.

3.5.6 Face validity method

The face validity method consisted of comparing resistance probabilities from the ensemble-calibrated model with the reported resistance levels of published clinical studies performed in Kazakhstan. The antibiotic-pathogen combinations selected were those with explicit reporting of resistance percentages and clear-cut end-points for the antibiotics. The predicted probability was estimated by the mean value of the resistance probability for the particular pathogen-antibiotic combination within the span of the years under consideration. Deviations of more than ±10 percent were considered unacceptable and analyzed with respect to directionality. Given the 9 small sample size (n=9), no formal statistical test was applied; comparisons are interpreted directionally.

3.6 Data and Code Availability

All the source code, pre-trained models, isotonic calibration artifacts, and macro-feature inference dataset for Kazakhstan are available at Zenodo (`https://doi.org/10.5281/zenodo.20230531`). The global isolate-level training data cannot be made freely available, as they are covered by Data Use Agreements with Vivli.org (ATLAS: Pfizer, 2023; SIDERO-WT: Shionogi, 2023; KEYSTONE). Reproducing the complete training pipeline requires separate data access applications through Vivli.org.

4 Results

4.1 Data Overview

The final dataset comprised 20,446,868 binary-coded isolate records from 85 countries (2015–2023). The resistant isolates constituted 18.3% of the whole dataset (3,737,951 out of 20,446,868 records), resulting in a 1:4.5 ratio imbalance between the "resistant" and "susceptible" classes. In the time split, there were 11,207,215 training records (≤ 2018, 17.5% resistant), 3,607,997 validation records (2019–2020, 19.3% resistant), and 5,631,656 hold-out test records (> 2020, 19.2% resistant) (Table 5).

Table 5: *Dataset split by time period with class distribution.*

Data Set	**Total**	**Resistant**	**Susceptible**
Train Set (≤2018)	11,207,215	1,960,713 (17.5%)	9,246,502 (82.5%)
Validation set (2019-2020)	3,607,997	695,943 (19.3%)	2,912,054 (80.7%)
Test set (>2020)	5,631,656	1,081,295 (19.2%)	4,550,361 (80.8%)
Total Dataset	20,446,868	3,737,951 (18.3%)	16,708,917 (81.7%)

4.2 Model Performance

In terms of performance on the holdout dataset (>2020), AutoGluon had the best AUC (0.8590) and AUPRC (0.6521), but the poorest recall among top-performing models (0.3977). Although LightGBM achieved marginally higher recall (0.7826 vs. 0.7761), CatBoost was selected as the final model because, among the high-recall models, it attained the best AUPRC (0.6464 vs. 0.6203) and AUC (0.8558 vs. 0.8543), satisfying the pre-specified recall-and-AUPRC criterion (Section 2.5.1). CatBoost's test dataset values for AUC, AUPRC, and F1 scores were 0.8558, 0.6464, and 0.5428, respectively (Table 6). By the DeLong test, the AUC for AutoGluon was significantly better than CatBoost

($p < 0.001$) and LightGBM ($p = 0.011$); however, no significant difference was found between CatBoost and LightGBM ($p > 0.99$). CatBoost, AutoGluon, and LightGBM performed significantly better than other models ($p < 0.001$; Supplementary Table S6). The pre-calibration BSS of -0.0162 reflects the systematic upward probability shift induced by the scale_pos_weight parameter, which is an expected trade-off when optimizing for recall under class imbalance; isotonic regression corrected this without affecting AUC. The pre-calibration ECE was 0.1780, which was reduced to 0.0122 following isotonic regression (Table 7).

Table 6: *Models' performances on the hold-out test set (>2020).*

Model	**AUPRC**	**AUC**	**F1-Score**	**Precision**	**Recall**	**Brier**	**BSS**
AutoGluon	0.6521	0.8590	0.5164	0.7361	0.3977	0.1049	0.3236
CatBoost	0.6464	0.8558	0.5428	0.4174	0.7761	0.1577	-0.0162
LightGBM	0.6203	0.8543	0.5458	0.4190	0.7826	0.1605	-0.0349
Random Forest	0.3907	0.7104	0.3912	0.4293	0.3593	0.1588	-0.0235
MLP	0.3769	0.7133	0.4064	0.2757	0.7729	0.2739	-0.7658
XGBoost	0.3667	0.6647	0.2990	0.4744	0.2183	0.1644	-0.0598
NGBoost	0.3643	0.6764	0.3892	0.3846	0.3939	0.1764	-0.1373
Logistic Regression	0.3386	0.6698	0.3912	0.3152	0.5153	0.1984	-0.2789

Table 7: *Expected Calibration Error before and after isotonic regression for all models.*

Model	**Val N**	**ECE before**	**ECE after**	**ΔECE**	**AUC before**	**AUC after**
Random Forest	3,607,997	0.1193	0.0056	-0.1137	0.7104	0.7111
NGBoost	3,607,997	0.1588	0.0058	-0.1530	0.6764	0.6863
XGBoost	3,607,997	0.1316	0.0068	-0.1248	0.6647	0.6654
Logistic Regression	3,607,997	0.2179	0.0071	-0.2108	0.6698	0.6696
AutoGluon	3,607,997	0.0149	0.0121	-0.0028	0.8590	0.8590
CatBoost	3,607,997	0.1780	0.0122	-0.1658	0.8558	0.8558
LightGBM	3,607,997	0.1793	0.0191	-0.1602	0.8543	0.8543
MLP	3,607,997	0.3254	0.0740	-0.2514	0.7133	0.7063

Using five-fold rolling time-based cross-validation yielded a mean AUC score of 0.8868 ± 0.0072, a mean AUPRC score of 0.6750 ± 0.0222, and a mean F1 score of 0.5636 ± 0.0159 (Table 8). This is a higher AUC score than that of the independent hold-out test set (0.8868 vs. 0.8558).

The confusion matrix is presented in Figure 2a.

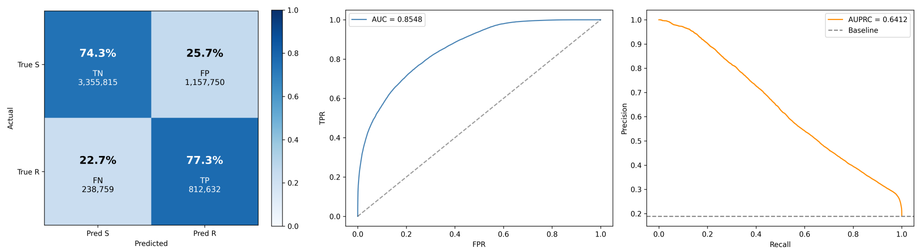

Figure 2: *Model performance evaluation on the hold-out test set (or temporal validation). (a) Confusion matrix of the CatBoost model showing the distribution of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). (b) Receiver Operating Characteristic (ROC) curve with Area Under the Curve (AUC). (c) Area Under the Precision-Recall Curve (AUPRC) of the CatBoost model*

Table 8: *Five-fold time-aware cross-validation results for CatBoost.*

Fold / Period	**AUC**	**AUPRC**	**F1 Score**
Fold 2015 → 2016	0.8867	0.6431	0.5412
Fold 2016 → 2017	0.8954	0.6761	0.5609
Fold 2017 → 2018	0.8759	0.6649	0.5587
Fold 2018 → 2019	0.8902	0.6918	0.5747
Fold 2019 → 2020	0.8859	0.6988	0.5824
Mean (± SD)	0.8868 ± 0.0072	0.6750 ± 0.0222	0.5636 ± 0.0159

4.3 Feature Analysis

Organism (34.7%) and antibiotic type (31.1%) were the most important features, contributing 65.8% of all predictive power (Table 9). The most important macro-feature was per-class antibiotic consumption (log consumption DDD per class), which had a value of 7.3%. The results of the ablation experiment indicated that no single feature group contributed substantially to model performance because removing a feature could only reduce AUC by a maximum of 0.0011 (ablation evaluated on the validation set; see Table 10 footnote). The SHAP plots (Figure 3) showed that log consumption DDD per class had SHAP values concentrated near zero with a modest positive tendency, while high values of log GDP per capita and health expenditure pct GDP were associated with negative SHAP contributions, indicating that wealthier and higher-spending countries tend toward lower predicted resistance. The divergence between the two ranking methods for temporal features such as year follows from the methodological difference described in Section 2.5.5.

Table 9: *CatBoost feature importance scores (PredictionValuesChange).*

Feature	**Importance Score (%)**
organism	34.6558
antibiotic	31.1463
antibiotic_class	7.6912
log_consumption_did_per_class	7.3168
year	4.2209
log_gdp_per_capita	2.8709
health_expenditure_pct_gdp	2.6757
population_density_per_sq_km	1.6725
sanitation_access_pct	1.6521
mean_temp_celsius	1.5660
log_consumption_ddd	1.5185
water_access_pct	1.4569
hospital_beds_per_1000	1.1803
aware_access_pct	0.3760

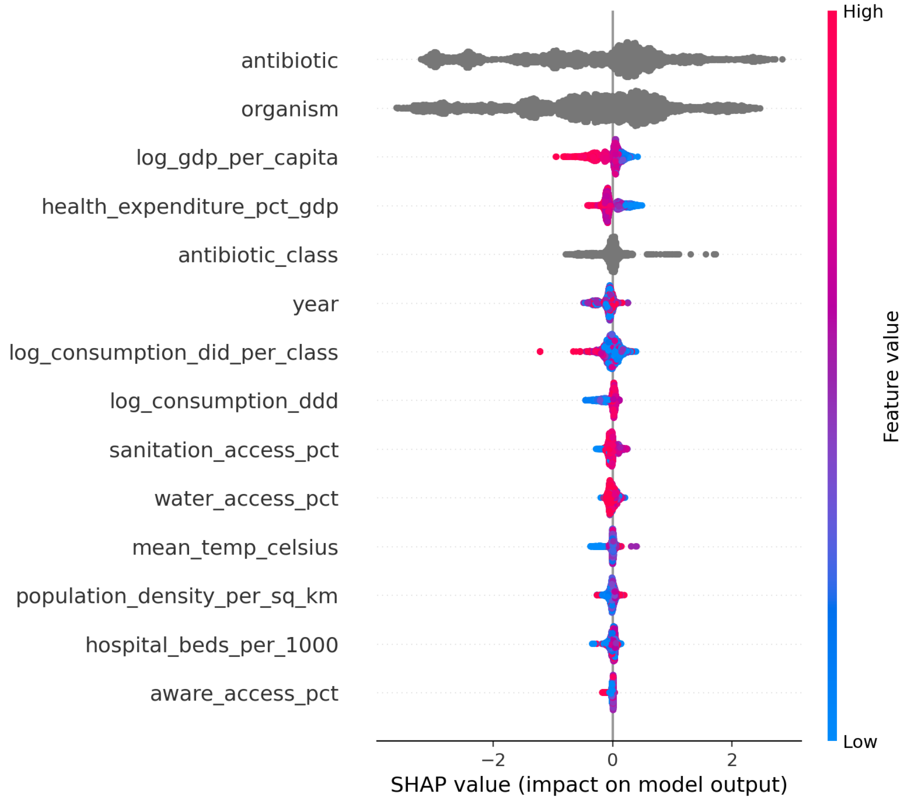

Figure 3: *SHAP beeswarm plot showing mean absolute feature contributions to model predictions (n = 5,000 stratified test samples).*

Table 10: *Ablation study results evaluated on five-fold cross-validation (2015–2020)*

Feature Set	**AUC**	**AUPRC**	**F1 Score**
All Features (Baseline)	0.8762	0.6814	0.5669
Drop antibiotic_class	0.8757	0.6793	0.5671
Drop Macro (gdp, health)	0.8755	0.6808	0.5604
Drop Consumption	0.8756	0.6815	0.5672
Drop Infrastructure (sanitation, water, beds)	0.8751	0.6779	0.5669
Drop Climate (temp, aware)	0.8765	0.6816	0.5710

4.4 Validation Results

Results of cluster-based validation and LOCO validation are provided in Table 11. The post-Soviet classifier performed with the best validation AUC

(0.8983) despite the use of just 2.5% of all training data (Table 11), while calibration by isotonic regression improved the Brier score from 0.1408 to 0.1161 without affecting the classifier's discriminative ability. The structural peers classifier (AUC = 0.8781) did not classify any Kazakhstan pathogens into the resistant class following calibration because of the domain mismatch described in Section 4.4.2. In country hold-out testing (Table 12), isotonic calibration had mixed effects on probability accuracy: the Brier score improved for France (0.137 to 0.079) and marginally for Turkey (0.149 to 0.147) but worsened for India (0.186 to 0.210) and Russia (0.142 to 0.151), indicating that calibration transfer was not uniformly reliable across countries.

Two-model ensemble prediction with equal weighting resulted in the average probability of resistance in Kazakhstan pathogens being 0.3215 among 1,098 prediction cases, with 54.6% of predictions exceeding the clinically significant threshold of 20%. Sensitivity analysis of the three weighting methods demonstrated consistent identification of the high-risk pathogens and direction of results (Table 13).

Table 11: *Performance of post-Soviet, structural peers, and global LOCO models, including Kazakhstan inference statistics.*

Metric Category	**Feature**	**Post-Soviet**	**Structural Peers**	**All Except KZ**
Dataset Info	Train Rows (≤2018)	282,170	758,387	11,207,215
	Val Rows (2019-20)	163,657	315,152	3,607,997
	Pos. Weight Scale	3.0026	3.5921	4.7159
Training	Best Iteration	644	497	3,173
	Best Test Score	0.8972	0.8789	0.8889
Validation	AUC	0.8983	0.8781	0.8890
(2019-2020)	AUPRC	0.8046	0.7226	0.6991
	Brier (Raw)	0.1408	0.1575	0.1431
Calibration	Brier (Calibrated)	0.1161	0.1184	0.0963
KZ Predictions	Mean Prob (Raw)	0.5343	0.4700	0.5059
	Mean Prob (Cal)	0.3657	0.2242	0.2772
	Predicted Resistance	34.0%	0.0%	19.3%

Table 12: *Country hold-out evaluation for Russia, Turkey, France, and India.*

Country (Metadata: Train, Test, SPW)	**Type**	**AUC**	**AUPRC**	**Recall**	**Prec.**	**F1**	**Brier**
India 11.18M, 206k, 4.74	Raw	0.808	0.765	0.785	0.622	0.694	0.186
	Calibrated	0.809	0.762	0.376	0.804	0.512	0.210
France 10.19M, 1.76M, 4.60	Raw	0.882	0.624	0.813	0.353	0.492	0.137
	Calibrated	0.882	0.616	0.378	0.748	0.502	0.079
Turkey 11.08M, 206k, 4.74	Raw	0.847	0.691	0.698	0.561	0.622	0.149
	Calibrated	0.847	0.685	0.288	0.857	0.431	0.147
Russia 11.05M, 193k, 4.75	Raw	0.855	0.649	0.652	0.564	0.605	0.142
	Calibrated	0.854	0.644	0.228	0.730	0.347	0.151

Table 13: *Sensitivity analysis of ensemble weighting schemes on isotonic-calibrated Kazakhstan predictions (n = 1,098). Where PS = Post Soviet; SP = Structural Peers; LOCO = All countries except Kazakhstan.*

Weighting Scheme	**Mean Probability**	**% > 0.20**
Two-model equal: PS + LOCO (primary)	0.3215	54.64
Three-model equal — PS + SP + LOCO	0.2892	62.48
Three-model weighted — 0.45 PS + 0.25 SP + 0.30 LOCO	0.3039	62.48

4.5 Kazakhstan Resistance Inference

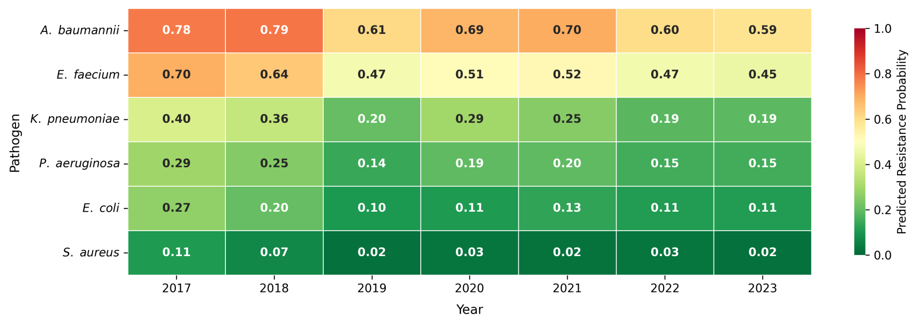

Figure 4: *Mean ensemble resistance probabilities for six WHO-priority pathogens in Kazakhstan, 2017–2023. Colour intensity encodes probability from 0 (dark green) to 1.0 (dark red).*

Table 14: *Mean ensemble-predicted resistance probabilities (2017–2023) with 95% bootstrap CIs (1,000 resamples) for six WHO-priority pathogens in Kazakhstan. CIs reflect ensemble consistency, not accuracy. ↓* denotes FDR significance at* $p_{\mathrm{BH}} < 0.05$*; Bonferroni-adjusted values > 1.0 are reported as > 0.99.*

Organism	Mean	95% CI	2017	2023	Trend	ρ	p (BH)	p (Bonf)
A. baumannii	0.69	[0.68, 0.70]	0.78	0.59	↓*	−0.82	0.047	0.141
E. faecium	0.55	[0.54, 0.57]	0.70	0.45	↓*	−0.86	0.041	0.082
K. pneumoniae	0.28	[0.26, 0.30]	0.40	0.19	↓*	−0.86	0.041	0.082
P. aeruginosa	0.20	[0.19, 0.21]	0.29	0.15	→	−0.61	0.222	0.889
E. coli	0.16	[0.15, 0.17]	0.27	0.11	→	−0.39	0.383	> 0.99
S. aureus	0.05	[0.04, 0.05]	0.11	0.02	→	−0.54	0.258	> 0.99

Figure 4 shows the average calibrated resistance probabilities (prob primary) for the six WHO priority pathogens in Kazakhstan from 2017 to 2023. These probabilities were computed using a two-model ensemble and were averaged for all antibiotics mapped for each organism per year. Table 14 presents these probabilities over the entire duration of the study together with their 95% bootstrap confidence intervals (1,000 resamples). The results for each antibiotic-pathogen combination are provided in Supplementary Table S5.

The pathogen with the highest mean resistance probability was *A. baumannii* (mean 0.69; range 0.59–0.78). This was followed by *E. faecium* (mean 0.55; range 0.40–0.70) and *K. pneumoniae* (mean 0.28; range 0.19–0.40). *S. aureus* had the lowest predicted resistance (mean 0.05; 95% CI: 0.04–0.05). Generally, in 2017–2018, predicted probabilities were highest, and then they decreased for all organisms before 2023. The annual mean probability series (2017–2023) for each organism was subjected to Spearman rank correlation tests, with 6 simultaneous tests carried out using the Benjamini-Hochberg FDR correction. There are three organisms for which temporal trends with FDR-adjusted significance were identified, which should be considered hypothesis-generating signals rather than confirmed trends: *E. faecium* ($\rho = -0.86$, $p_{\mathrm{BH}} = 0.041$), *K. pneumoniae* ($\rho = -0.86$, $p_{\mathrm{BH}} = 0.041$), and *A. baumannii* ($\rho = -0.82$, $p_{\mathrm{BH}} = 0.047$). No trends were found for the remaining three pathogens ($p_{\mathrm{BH}} \geq 0.222$).

The highest resistance probabilities by antibiotic class were: carbapenems and fluoroquinolones for *A. baumannii*; glycopeptides for *E. faecium*; fluoroquinolones for *E. coli*; cephalosporins for *K. pneumoniae*; and aminoglycosides and carbapenems for *P. aeruginosa*.

5 Discussion

5.1 Key Findings

Our hypothesis is that the underlying factors determining AMR are universal; this hypothesis was tested using a zero-shot approach. That is indeed the hypothesis that is supported by a global LOCO AUC of 0.8558—without any Kazakhstani data to train with—although that does not constitute proof as no Kazakhstani data exist with which to directly measure the performance. The most important feature was organism and antibiotic identity (65.8%; Table 9), suggesting that resistance ranking is primarily a universal property of the organism–antibiotic pair. This result is in addition to MacFadden et al. (2018) where such ranking was recovered using isolates collected locally. The high discriminative performance in the absence of country data indicates that the ranking order is more related to the identity of the organism/antibiotic than to country. The macro-feature set (variables of consumption, macroeconomic, and population) on the other hand defines the absolute resistance (calibration), thus allowing daily prediction in the period 2017–2023. As there were not any isolates provided from Kazakhstan, the macro-driven signals are the only channel through which country-level variation enters the temporal trend analysis. The difference in the model means for the country level "Kazakhstan" is presented in Table 11: the model means vary significantly (0.366, 0.224, 0.277) for the vector "Kazakhstan". The macro-signal sets the baseline probability, and it is the level (not the rank) that is nearly equal in all models. We constructed a post-Soviet cluster model that was designed from just 2.5% of the training data, and achieved an AUC of 0.8983 to show that the post-Soviet legacy of shared healthcare and shared antibiotic use is very useful for the prior information. The likelihood of resistance in Kazakhstan, therefore, is more similar to that in post-Soviet countries than in the rest of the world. Resistance was predicted at the highest level for *A. baumannii* (mean 0.69) and *E. faecium* (mean 0.55). Three pathogens showed moderate to high declines after FDR correction: *A. baumannii* ($\rho = -0.82$, $p_{\mathrm{BH}} = 0.047$), *E. faecium* ($\rho = -0.86$, $p_{\mathrm{BH}} = 0.041$), and *K. pneumoniae* ($\rho = -0.86$, $p_{\mathrm{BH}} = 0.041$). These declining trends are consistent with the decreased national use of Watch-class antibiotics in 2017–2023 (Semenova et al., 2024). High absolute resistance persisting despite a downward trend likely reflects intrinsic mechanisms beyond consumption-driven selection. SHAP analysis associated higher GDP and health expenditure with lower predicted resistance, consistent with the development–resistance relationship described by Laxminarayan et al. (2013).

5.2 Model Selection

The model selection process considered clinical rather than solely statistical factors. Although the AUC of AutoGluon was slightly greater (0.8590 compared to 0.8558), the recall performance of CatBoost was much better (0.7761 compared to 0.3977). The poor recall of AutoGluon against the class imbalance can be considered a drawback of AUC optimization in an automated machine learning context (He & Garcia, 2009). For the clinical relevance of AMR surveillance, the omission of resistant strains is clinically more important than a possible over-prediction of cases; accordingly, recall is the primary clinical endpoint, with AUPRC as co-primary. Even though it may have been possible to improve the recall of AutoGluon through reducing the classification threshold, this optimization could only be performed based on locally collected data, which were not available for Kazakhstan; hence, the superiority of CatBoost's recall at the default setting provides a realistic comparison. Because the framework is intended to prioritize testing rather than to replace measurement, high recall is the appropriate objective: the cost of overlooking a high-risk combination that should have been tested outweighs the cost of a false alarm.

The isotonic regression for calibration of CatBoost improved ECE from 0.1780 to 0.0122 but did not reduce the discriminatory power of the model. The negative BSS of -0.0162 before the isotonic regression is expected because CatBoost was calibrated using the scale_pos_weight hyperparameter to achieve higher recall; model predictions have inherent bias towards predicting positives and cannot be considered well-calibrated probabilities without isotonic regression. On the other hand, the method of structural peers was not used as part of the main study results because the application of isotonic regression to the high-income countries (Qatar, Turkey, Romania, Bulgaria, Hungary, and Poland) resulted in the Kazakhstan predictions going below a clinically relevant point, which can only be solved with the use of Kazakhstan-specific calibration data. There was no sensitivity analysis that contradicted this decision. The ultimate drop in performance of the structural peers model to 0% predicted resistance is noteworthy in itself: macroeconomic clustering is insufficient as a basis for selecting peers according to AMR prediction requirements, presumably because the legacy of the healthcare system and past antibiotic stewardship policy play more significant roles in forming the pattern of resistance as compared to GDP and population density. It follows that the reason for adopting a prespecified two-model primary ensemble made up of the post-Soviet and global LOCO models lies in the above weaknesses, as the latter is better fitted to the epidemiological context of Kazakhstan. The structural peers model is included in the ensemble just as an additional sensitivity analysis option. The cutoff rate adopted (20%) is justified based on the previous research in the realm of antibiotic stewardship policy (Gupta et al., 2011; Kollef et al., 2021; WHO, 2019).

5.3 Comparison with Existing Literature

Prior approaches relied on ML methods applied to genomic data (Moradigaravand et al., 2018; Drouin et al., 2019), thus being unfeasible in situations where the sequencing centers were unavailable. MacFadden et al. (2018) demonstrated that environmental and population-level factors such as ambient temperature are associated with AMR, but their analysis relied on regional surveillance data from a single country. As opposed to previous approaches, the approach presented herein uses macro-proxy data to produce AMR predictions for all six priority pathogens, alongside several antibiotics, while omitting the use of any locally-derived Kazakhstani strains.

In terms of discriminative performance, the multi-strain XGBoost algorithm trained on the same ATLAS database but not constrained by zero-shot limitations had an AUC of 0.96 (Valavarasu et al., 2025); species-specific MALDI-TOF models scored between 0.81 and 0.85 (Wiesmann et al., 2025); genome-based NN models scored between 0.83 and 0.94 (Aytan-Aktug et al., 2020); while the patient-level EHR models scored 0.80 (Kim et al., 2026). Therefore, the AUC of 0.8558 under zero-shot conditions is comparable to the genomic models that are developed for each individual drug, and is better than the models developed at the patient level.

5.4 Interpretation of Prediction Results of Resistance in Kazakhstan

With model performance established, we turn to resistance inference for Kazakhstan.

5.4.1 Summary of Results

The clinically actionable layer of the inference results is given by the antibiotic class breakdown (Supplementary Table S5). The model also flags carbapenems and fluoroquinolones as the most resistant classes for *A. baumannii*, consistent with the MDR that has been reported in clinical settings in post-Soviet countries (Semenova et al., 2024). Elevated glycopeptide resistance probabilities were observed for *E. faecium*, for which vancomycin is a last-resort agent. The overall probabilities of resistance to cephalosporins were higher than those for any other antibiotic for *K. pneumoniae* (mean 0.28), and *P. aeruginosa* (mean 0.20) was at the clinical threshold with higher probabilities of resistance to aminoglycosides and carbapenems. Average prevalence levels of *E. coli* and *S. aureus* are below threshold limits, though fluoroquinolone resistance in *E. coli* warrants monitoring. Temporal, or time-based, trends are discussed for the three highest-risk pathogens in Section 4.1.

5.4.2 Calibration Artefact

The structural (ST) peers model (Qatar, Turkey, Romania, Bulgaria, Hungary, Poland) yielded 0.0% predicted resistance after isotonic calibration, as the mapping from raw to estimated probabilities during calibration produced compression of all raw probabilities to below the clinical threshold (calibrated mean = 0.22; Supplementary Figure S1). This failure is valuable insofar as it illustrates that while a degree of macroeconomic similarity appears necessary, it is not enough to foster a degree of peer selection; Kazakhstan shares GDP profiles with these structural peers but differs in its post-Soviet antibiotic stewardship legacy. That is why the "structural peers" are included as a sensitivity check only, instead of being part of the primary ensemble. A sensitivity analysis (Table 13) shows that none of the three weighting option scenarios, which include the incorporation of the structural peers model, drive the main findings; all three weighting schemes identified more than half of inference cases exceeding the 20% threshold, confirming that the exclusion of the structural peers model does not drive the main findings.

5.4.3 Face Validity

Face validity was assessed by comparing ensemble-calibrated resistance probabilities against observed resistance rates from Strochkov et al. (2025) for nine *A. baumannii* isolates collected from pneumonia patients at three clinical facilities in Karaganda, Kazakhstan (2022–2023), as described in Section 2.5.

Carbapenem resistance (meropenem/imipenem) was predicted at 73.6%, compared to an observed 66.67% ($\Delta = +6.9$pp), falling within the ±10pp acceptance threshold. Amikacin resistance was predicted at 69.0%, within the threshold from an observed 77.78%. Gentamicin resistance was under-predicted by 24.0pp (predicted 64.9%, observed 88.89%). Cephalosporin resistance was underestimated by 26.4pp compared to an observed 100%. Fluoroquinolone class resistance was reported with wide uncertainty (95% CI: 70.09–100%); using the point estimate of $\sim$100%, the model underestimated by around 30pp. The under-prediction for cephalosporin and fluoroquinolone may be due to the high level of resistance of MDR isolates selected in a tertiary hospital setting, which may not be fully captured by consumption-based features used at the national level.

The comparisons suggest that the model provides reasonable predictions of resistance for antibiotic classes under sustained national selective pressure (carbapenems, aminoglycosides), but it underestimates resistance for those classes which are more strongly influenced by hospital-specific selective pressure. This comparison has a small sample size (n=9) and is from a single city, so it is not considered generalisable but is an indicator, pending validation with larger numbers.

Table 15: *Face validity comparison of predicted versus observed A. baumannii resistance rates from Strochkov et al. (2025); n=9 isolates, pneumonia inpatients, Karaganda, Kazakhstan (2022–2023). All comparisons are for pneumonia inpatients in Karaganda (2022–2023). Point estimate (∼100%) used for Δ calculation.*

Organism	**Antibiotic**	**Observed %**	**Predicted %**	**Δ**	**Direction**
A. baumannii	Meropenem/Imipenem	66.67	73.6	+6.9	Within 10%
A. baumannii	Amikacin	77.78	69.0	−8.8	Within 10%
A. baumannii	Gentamicin	88.89	64.9	−24.0	Under-predicts
A. baumannii	Cephalosporins (class)	100.00	73.6	−26.4	Under-predicts
A. baumannii	Fluoroquinolones (class)	∼100.00	70.0	−30.0	Under-predicts

5.5 Limitations

The main drawback of the study is the lack of isolate data from Kazakhstan: the model has not been trained on samples from Kazakhstan, and all predictions are provisional and must be validated in clinical practice in the future. The prediction grid does not contain the three locally used antibiotics—furazidin, nitroxoline, and roxithromycin—which have no international equivalents. Excluding intermediate resistance categories could also lead to an underestimation of partial resistance.

Face validity against nine *A. baumannii* isolates from Karaganda showed acceptable predictions for carbapenems and amikacin, which are subject to sustained national selection pressure, but systematic under-prediction for hospital-dominated classes; full details are in Section 4.4.3. There were no similar previously published data for *K. pneumoniae* or *P. aeruginosa* in Kazakhstan.

Calibration was performed using data from high-income countries, which might not properly represent the Kazakhstani resistance landscape. The confidence intervals in Table 14 reflect ensemble consistency, not validated accuracy, and should not be interpreted as true resistance prevalence estimates. Because of the small sample sizes involved, there are only a few observations per year for the temporal trends, and the p_{BH} of 0.041 to 0.047 suggest trends that may not be considered real trends given the small number of observations per year, and should be interpreted as a possible indicator of a trend which is likely to be confirmed with a larger number of observations. Hyperparameters were also optimized using the whole dataset instead of specific data folds, thereby increasing the LOCO-AUC estimates by a small margin, but this effect is expected to be minimal, as no single country contributes more than 5% of the data for a single LOCO fold (Roberts et al., 2017).

5.6 Future Directions

The next step is to test 100–200 local isolates of the predicted high-risk species (*A. baumannii* and *E. faecium*) in a pilot validation study (2025–2026). The threshold used in Section 4.4.3 was applied here, and a positive result would be defined as predictions being within ±10pp of the observed rates for the majority of the predictions. The method is easily adaptable to countries that lack surveillance data but have good quality data on macroeconomic performance and antibiotic usage, as is the case in much of Central Asia, sub-Saharan Africa, and Southeast Asia.

6 Conclusion

Since no local clinical data were available, we used zero-shot transfer learning to predict antimicrobial resistance probabilities in Kazakhstan using global consumption and surveillance data and macroeconomic data. The CatBoost model, trained on 20.4 million isolates from 85 countries, had an AUC of 0.8558, an AUPRC of 0.6464, and a recall of 0.7761 on the hold-out test set. A cluster model using only 2.5% of the data matched or exceeded the global model for Kazakhstan, indicating that shared Soviet healthcare heritage provides stronger prior information for Kazakhstan than the global average. The calibrated two-model ensemble identified *A. baumannii* and *E. faecium* as the highest-risk pathogens, with three of six pathogens (*A. baumannii*, *E. faecium*, and *K. pneumoniae*) showing declining trends, which is consistent with the decrease in Watch-class antibiotic usage in Kazakhstan. Face-validity comparison of predictions was supported for antibiotics under national selection pressure, but the results are preliminary due to lack of Kazakhstan isolates for training and testing the model and calibration performed using high-income country data. The framework, in this sense, should be viewed as hypothesis-generating and not as a substitute for laboratory-based surveillance, as it focuses on which organism–antibiotic combinations should be validated first.

References

Aytan-Aktug, D., Clausen, P. T. L. C., Bortolaia, V., Aarestrup, F. M., & Lund, O. (2020). Prediction of acquired antimicrobial resistance for multiple bacterial species using neural networks. *mSystems, 5*(1), Article e00774-19. https://doi.org/10.1128/mSystems.00774-19

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). https://doi.org/10.1145/2939672.2939785

Cox, D. R. (1958). The regression analysis of binary sequences. *Journal of the Royal Statistical Society: Series B (Methodological), 20*(2), 215–

242.

DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics, 44*(3), 837–845.

Drouin, A., Letarte, G., Raymond, F., Marchand, M., Corbeil, J., & Laviolette, F. (2019). Interpretable genotype-to-phenotype classifiers with performance guarantees. *Scientific Reports, 9*, Article 4071. https://doi.org/10.1038/s41598-019-40561-2

Duan, T., Avati, A., Ding, D. Y., Thai, K. K., Basu, S., Ng, A. Y., & Schuler, A. (2020). NGBoost: Natural gradient boosting for probabilistic prediction. In *Proceedings of the 37th International Conference on Machine Learning* (PMLR, Vol. 119, pp. 2690–2700).

Erickson, N., Mueller, J., Shirkov, A., Zhang, H., Larroy, P., Li, M., & Smola, A. (2020). *AutoGluon-Tabular: Robust and accurate AutoML for structured data* (arXiv:2003.06505). https://doi.org/10.48550/arXiv.2003.06505

European Committee on Antimicrobial Susceptibility Testing. (2023). *Breakpoint tables for interpretation of MICs and zone diameters, version 13.0.* http://www.eucast.org

Gupta, K., Hooton, T. M., Naber, K. G., Wullt, B., Colgan, R., Miller, L. G., Moran, G. J., Nicolle, L. E., Raz, R., Schaeffer, A. J., & Soper, D. E. (2011). International clinical practice guidelines for the treatment of acute uncomplicated cystitis and pyelonephritis in women: A 2010 update by the Infectious Diseases Society of America and the European Society for Microbiology and Infectious Diseases. *Clinical Infectious Diseases, 52*(5), e103–e120. https://doi.org/10.1093/cid/ciq257

He, H., & Garcia, E. A. (2009). Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering, 21*(9), 1263–1284. https://doi.org/10.1109/TKDE.2008.239

Kassym, L., Kussainova, A., Semenov, D., Aimurziyeva, A., Uzbekova, S., & Semenova, Y. (2025). National trends in azithromycin consumption during 2017–2023 in Kazakhstan: Impact of the COVID-19 pandemic and the imperative for enhanced clinical guidelines. *Scientific Reports, 15*, Article 6309. https://doi.org/10.1038/s41598-025-91216-4

Ke, G., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Proceedings of the 31st Conference on Neural Information Processing Systems (NIPS 2017)*, Long Beach, 4–9 December 2017, 3149–3157. https://doi.org/10.5555/3294996.3295074

Kim, Y., Jeong, I., Park, J.-H., et al. (2026). Multi-task learning based early prediction model for antibiotic resistance using multi-institutional cohort data. *Scientific Reports, 16*, Article 11891. https://doi.org/10.1038/s41598-026-41185-z

Kollef, M. H., Shorr, A. F., Bassetti, M., Timsit, J.-F., Micek, S. T., Michelson, A. P., & Garnacho-Montero, J. (2021). Timing of antibiotic therapy in the ICU. *Critical Care, 25*(1), Article 360. https://doi.org/10.1186/s13054-021-03787-z

Kosherova, Z., Zhazykhbayeva, D., Aimurziyeva, A., Bayesheva, D., & Semenova, Y. (2025). The implementation of antimicrobial consumption surveillance and stewardship in human healthcare in post-Soviet states: A systematic review. *Antibiotics, 14*(8), Article 749. https://doi.org/10.3390/antibiotics14080749

Laxminarayan, R., Duse, A., Wattal, C., Zaidi, A. K. M., Wertheim, H. F. L., Sumpradit, N., Vlieghe, E., Hara, G. L., Gould, I. M., Goossens, H., Greko, C., So, A. D., Bigdeli, M., Tomson, G., Woodhouse, W., Ombaka, E., Peralta, A. Q., Qamar, F. N., Mir, F., . . . Cars, O. (2013). Antibiotic resistance — The need for global solutions. *The Lancet Infectious Diseases, 13*(12), 1057–1098. https://doi.org/10.1016/S1473-3099(13)70318-9

Lim, L., Kassym, L., Kussainova, A., Aubakirova, B., Semenova, Y., & Jakupi, A. (2025). Optimizing antibiotic stewardship and reducing antimicrobial resistance in Central Asia: A study protocol for evidence-based practice and policy. *PLoS ONE, 20*(1), Article e0307784. https://doi.org/10.1371/journal.pone.0307784

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*, 4765–4774.

MacFadden, D. R., McGough, S. F., Fisman, D., Santillana, M., & Brownstein, J. S. (2018). Antibiotic resistance increases with local temperature. *Nature Climate Change, 8*(6), 510–514. https://doi.org/10.1038/s41558-018-0161-6

Moradigaravand, D., Palm, M., Farewell, A., Mustonen, V., Warringer, J., & Parts, L. (2018). Prediction of antibiotic resistance in *Escherichia coli* from large-scale pan-genome data. *PLOS Computational Biology, 14*(12), Article e1006258. https://doi.org/10.1371/journal.pcbi.1006258

Naghavi, M., Vollset, S. E., Ikuta, K. S., et al. [GBD 2021 Antimicrobial Resistance Collaborators]. (2024). Global burden of bacterial antimicrobial resistance 1990–2021: A systematic analysis with forecasts to 2050. *The Lancet, 404*(10459), 1199–1226. https://doi.org/10.1016/S0140-6736(24)01867-1

Nori, H., Jenkins, S., Koch, P., & Caruana, R. (2019). *InterpretML: A uni-*

fied framework for machine learning interpretability (arXiv:1909.09223). https://doi.org/10.48550/arXiv.1909.09223

O'Neill, J. (2016). *Tackling drug-resistant infections globally: Final report and recommendations.* Review on Antimicrobial Resistance. https://amr-review.org/

Pfizer. (2023). *ATLAS database* [Data set]. Vivli. https://vivli.org/

Pires, S. M., Duarte, A. S. R., & Hald, T. (2021). A review of machine learning applications for antimicrobial resistance prediction. *Clinical Microbiology and Infection, 27*(2), 166–173. https://doi.org/10.1016/j.cmi.2020.09.025

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems 31* (pp. 6639–6649).

Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., & Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography, 40*(8), 913–929. https://doi.org/10.1111/ecog.02881

Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature, 323*(6088), 533–536. https://doi.org/10.1038/323533a0

Semenova, Y., Yergaliyeva, A., Aimurziyeva, A., Manatova, A., Kuntuganova, A., Makalkina, L., Aldiyarova, N., Semenov, D., & Lim, L. (2024). A nationwide evaluation of antibiotic consumption in Kazakhstan from 2019 to 2023. *Antibiotics, 13*(12), Article 1123. https://doi.org/10.3390/antibiotics13121123

Shionogi. (2023). *SIDERO-WT database* [Data set]. Vivli. https://vivli.org/

Strochkov, V., Beloussov, V., Orkara, S., Lavrinenko, A., Solomadin, M., Yegorov, S., & Sandybayev, N. (2025). Genomic characterization of multidrug-resistant *Acinetobacter baumannii* in pneumonia patients in Kazakhstan. *Diagnostics, 15*(6), Article 704. https://doi.org/10.3390/diagnostics15060704

Valavarasu, S., Sangu, Y., & Mahapatra, T. (2025). Prediction of antibiotic resistance from antibiotic susceptibility testing results from surveillance data using machine learning. *Scientific Reports, 15*, Article 30509. https://doi.org/10.1038/s41598-025-14078-w

Wang, L., Chen, H., Zhang, Y., Tian, Y., Hu, X., Wu, J., Li, X., Jia, H., Wang, H., Yu, C., Yang, Y., Fang, L., & Song, H. (2025). Global antibiotic consumption and regional antimicrobial resistance, 2010–21: An analysis of pharmaceutical sales and antimicrobial resistance surveillance data. *Lancet Global Health, 13*(11), e1880–e1891. https://doi.org/10.1016/S2214-109X(25)00308-0

Wiesmann, N., Enders, D., Westendorf, A., Koch, R., & Schaumburg, F. (2025). Prediction of antimicrobial resistance from MALDI-TOF mass spectra using machine learning: A validation study. *Journal of Clinical Microbiology, 63*, Article e01186-25. https://doi.org/10.1128/jcm.01186-25

World Bank. (2017). *Drug-resistant infections: A threat to our economic future.* https://www.worldbank.org/en/topic/health/publication/drug-resistant-infections-a-threat-to-our-economic-future

World Bank. (2023). *World Development Indicators* [Data set]. https://data.worldbank.org/

World Health Organization. (2019). *Critically important antimicrobials for human medicine* (6th revision). https://www.who.int/publications/i/item/9789241515528

World Health Organization. (2023). *AWaRe classification of antibiotics.* https://www.who.int/publications/i/item/WHO-MHP-HPS-EML-2023.04

Zhussupova, G., Utepova, D., Orazova, G., Zhaldybayeva, S., Skvirskaya, G., & Tossekbayev, K. (2021). Evaluation of antibiotic use in Kazakhstan for the period 2017–2019 based on WHO Access, Watch and Reserve classification (AWaRe 2019). *Antibiotics, 10*(1), Article 58. https://doi.org/10.3390/antibiotics10010058